

Exercise 7

Consider the conformal prediction with $\mathcal{Y} = \mathbb{R}$ with a given model $\hat{f} : \mathcal{X} \rightarrow \mathcal{Y}$ and calibration points $(X_1, Y_1), \dots, (X_n, Y_n)$. Put

$$\hat{q}_L = \text{Quantile} \left(\left\{ \hat{f}(X_i) - Y_i \right\}_{i=1}^n ; \left(1 - \frac{\alpha}{2}\right) \left(1 + \frac{1}{n}\right) \right),$$

$$\hat{q}_R = \text{Quantile} \left(\left\{ Y_i - \hat{f}(X_i) \right\}_{i=1}^n ; \left(1 - \frac{\alpha}{2}\right) \left(1 + \frac{1}{n}\right) \right),$$

and

$$\mathcal{C}(X_{n+1}) = \left[\hat{f}(X_{n+1}) - \hat{q}_L, \hat{f}(X_{n+1}) + \hat{q}_R \right].$$

Prove that

$$P \{ Y_{n+1} \in \mathcal{C}(X_{n+1}) \} \geq 1 - \alpha$$

by showing both

$$P \left\{ Y_{n+1} < \hat{f}(X_{n+1}) - \hat{q}_L \right\} \leq \frac{\alpha}{2}$$

and

$$P \left\{ Y_{n+1} > \hat{f}(X_{n+1}) + \hat{q}_R \right\} \leq \frac{\alpha}{2},$$

assuming that $(X_1, Y_1), \dots, (X_{n+1}, Y_{n+1})$ is exchangeable.

Solution We have

$$P \left\{ Y_{n+1} < \hat{f}(X_{n+1}) - \hat{q}_L \right\} = P \left\{ \hat{f}(X_{n+1}) - Y_{n+1} > \hat{q}_L \right\} = 1 - P \left\{ \hat{f}(X_{n+1}) - Y_{n+1} \leq \hat{q}_L \right\} \leq 1 - \left(1 - \frac{\alpha}{2}\right) = \frac{\alpha}{2}$$

where the last inequality uses Theorem 3.2 in the textbook with score $S_i = \hat{f}(X_i) - Y_i$. This is legitimate because the data is assumed to be exchangeable and the score is trivially symmetric in case of split conformal prediction. The right hand side inequality is proved in exactly the same way but with score $S_i = Y_i - \hat{f}(X_i)$. Combine the two, we have the desired marginal coverage. The proof is complete. ////